MAHATMA GANDHI UNIVERSITY, KOTTAYAM
MGU-UGP (HONOURS)

FIRST SEMESTER EXAMINATION

(2024 ADMISION ONWARDS)

MG1DSCNBC100 – FUNDAMENTALS OF BIOCHEMISTRY

Duration: 2 hrs Maximum Marks: 50

Students should attempt at least one question from each course outcome to enhance their overall outcome attainability.

Each question is tagged with appropriate Bloom's Revised Taxonomy level (Remembering, Understanding, Applying, Analyzing, Evaluating and Creating) and the relevant course outcome (CO) number.

*Remember (K), Understand (U), Apply (A), Analyse (An), Evaluate (E), Create (C)

Part A

Multiple Choice Questions
Answer **All** Questions
Each question carries **1** mark

1. Which of the following is a monosaccharide? [K] [1]

A) Sucrose
B) Lactose
C) Glucose
D) Maltose

2. What type of polysaccharide is agar? [U] [1]

A) Homopolysaccharide
B) Heteropolysaccharide
C) Monosaccharide
D) Tetra saccharides

3. The protein present in Rice is …… [U] [2]
   A) Oryzenin
   B) Glutenin
   C) Glutelin
   D) Gliadin

4. Glycerol combines with fatty acid to give……. [K] [2]

   A) Mono acyl glycerol
   B) Triglyceride
   C) Glycerol
   D) Monoglyceride

5. Vitamin D is synthesized in the skin in response to: [K] [3]
   A) Dietary intake
   B) Cold temperature
   C) Sunlight
   D) Vitamin E

6. Which of the following describes the structure of vitamin E: [K] [3]
   A) Long chain fatty acid
   B) A tocopherol or tocotrienol ring
   C) Delta-tocopherol
   D) A phenolic compound

(6x1=6 marks)

Part B

Short Answer Questions
Answer Any **two** Questions
Each question carries **two** marks

7. State the differences between Ionic and Covalent bond. [U] [1]
8. Explain the classifications of monosaccharides based on the number of carbon atoms.[U] [1]
9. Compare fibrous proteins and globular proteins in terms of structure and solubility.[U] [2]
10. Explain the synthesis of cholecalciferol .[K] [3]

(2x2=4 marks)

Part C

Short Essay Questions

Answer Any **four** Questions

Each question carries **five** marks

11. Explain the importance of glycosidic bond in carbohydrates [U] [1]
12. Describe the classification of proteins based on composition and solubility [K] [2]
13. Explain the classification of fatty acids. [K] [2]
14. Explain the role of vitamin A in our body.[U] [3]
15. Describe the structure of proteins [U] [2]
16. Explain the types of chemical bond[K] [1]

(4x5=20 marks)

Part D

Long Essay Questions

Answer Any **two** Questions

Each question carries **ten** marks

17. Discuss the various properties of carbohydrates. [U] [1]
18. Describe the classification of proteins .[K] [2]
19. Discuss on the classification and properties of lipids [K] [2]
20. Explain the structure, properties and role of vitamin D. [U] [2]

(2x10=20 marks)